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Chemistry 12

Lab 3A: Precipitation Reactions

1 Purpose

To determine the identity of the precipitate in the following double replacement reaction:

$$3\,BaCl_2\,(aq) + Al_2(SO_4)_3\,(aq) \rightleftharpoons 3\,BaSO_4 + 2\,AlCl_3$$

2 Procedure

2.1 Pre-lab Work

1. In a microwell plate, combine 2 drops each of $BaCl_2$ and $Al_2(SO_4)_3$.
2. Write a balanced chemical equation for the reaction.
3. Design a procedure that will allow you to experimentally determine the chemical formula of the precipitate, without consulting the datasheet or theory.
4. Carry out your procedure and record observations.

2.2 Lab Work

Due to a scheduling conflict, I followed Talan's procedure, which is the following:

1. Repeat step 1 from the pre-lab work, thoroughly rinsing the plate between iterations, with the following combinations:
   - $BaCl_2$ + $Al(NO_3)_3$
   - $BaCl_2$ + Na_2SO_4
   - $FeCl_3$ + Na_2SO_4

3 Data/Observations

The reagents were mixed and the following observations were noted:

First Reagent	Second Reagent	Observation
$BaCl_2$	$Al_2(SO_4)_3$	Precipitate
$BaCl_2$	$Al(NO_3)_3$	No precipitate
$BaCl_2$	Na_2SO_4	Precipitate
$FeCl_3$	Na_2SO_4	No precipitate

Table 1: Observations of whether a precipitate formed.

4 Analysis

Recalling the reaction equation:

$$3\,BaCl_2\,(aq) + Al_2(SO_4)_3\,(aq) \rightleftharpoons 3\,BaSO_4 + 2\,AlCl_3$$

The identity of the precipitate can be either $BaSO_4$ or $AlCl_3$.

$AlCl_3$ can be immediately proved to not be the precipitate. In the second trial, the following reaction occurred:

$$3\,BaCl_2 + 2\,Al(NO_3)_3 \rightleftharpoons 3\,Ba(NO_3)_2 + 2\,AlCl_3$$

Because no precipitate formed, we know that both $Ba(NO_3)_2$ and $AlCl_3$ are soluble in water. Therefore, $AlCl_3$ cannot be the precipitate.

$BaSO_4$ can be proved to be the precipitate by observing the reactions and noting that a precipitate only forms when both Ba^{2+} and ${SO_4}^{2-}$ are present in the reagents. In the first and third trials, both are present, and both form precipitates. In the second trial, ${SO_4}^{2-}$ is absent, and no precipitate is formed. In the fourth trial, Ba^{2+} is absent, and no precipitate is formed. The absence of either leads to no precipitate forming, meaning that both ions must be present in the percipitate. Combining them, the precipitate can be deduced to be $BaSO_4$.

5 Conclusion

Recalling the reaction equation:

$$3\,BaCl_2\,(aq) + Al_2(SO_4)_3\,(aq) \rightleftharpoons 3\,BaSO_4 + 2\,AlCl_3$$

In section 4 we concluded that the precipitate in the reaction is $BaSO_4$ and not $AlCl_3$. When $AlCl_3$ was produced separately in a reaction with neither Ba^{2+} nor ${SO_4}^{2-}$, no precipitate formed. When either of the ions in $BaSO_4$ were absent in other trials, no precipitate formed either. Therefore $AlCl_3$ is not the precipitate, and $BaSO_4$ must be the precipitate